Updated on October 18, 2026

MENGFAN XU

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ACADEMIC APPOINTMENT

University of Massachusetts Amherst, United States *Sep. 2024 - Present*
Assistant Professor, Mechanical and Industrial Engineering
Adjunct Assistant Professor, Manning College of Information & Computer Sciences, Apr. 2025 - Present

EDUCATION

Northwestern University, United States *Aug. 2018 - Jul. 2024*
Ph.D. in Industrial Engineering and Management Sciences
Dissertation title: **"Multi-armed Bandit with Several Agents and Objectives"**

University of Science and Technology of China, China *Jun. 2018*
B.S. in Statistics

Yale University, United States *Jun. 2017 - Sep. 2017*
BioStats, Summer exchange program

RESEARCH INTERESTS

Online Learning, Multi-Agent Systems, Statistical Learning and Algorithm Design, Large Language Model Optimization, and Stochastic Processes

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern at LinkedIn *Jun. 2023 - Sep. 2023*
• Developed natural language understanding models and large language models (LLMs) for keywords extraction based on user queries in search AI team;
• Prompt engineering and data mining;
• Developed a workflow and pipeline to ultimately improve users' search experience

Algorithm Engineer at Didi Chuxing *2021*
• Developed two-staged algorithms that achieve over 25% improvement on baselines for pricing of multiple ride options at Didi;
• Developed the very first algorithm with causal forest for model-based route-based pricing of Didi Discount Express and iterated multiple versions with superior performance;
• Built data pipelines with Spark for processing billions of data and advancing feature engineering to support large-scale model training and distributional computations;
• Built a model pipeline for online recommender systems with causal neural networks;
• Conducted research on treatment effect estimation: proving the convergence of GCF for non-parametric DRF, delivering multiple presentations, and completing a paper accepted for oral presentation at KDD 2022

AWARDS AND HONORS

Silver Reviewer at machine learning conference ICML ***2026***

Meritorious Reviewer at INFORMS Journal on Computing ***2025***

Top Reviewer at machine learning conference NeurIPS ***2024***

Honorable Mention at RLC workshop ***2024***

Spotlight presentation at machine learning conference NeurIPS (top 3%) *2023*

Walter P. Murphy Graduate Fellowship, Northwestern University *2018 - 2019*

Runner up, Guo Moruo Scholarship, USTC (the highest honor) *2018*

National Scholarship, Chinese Ministry of Education (top 1%) *2017*

National Scholarship, Chinese Ministry of Education (top 1%) *2016*

Mobile Commerce Competition, USTC (top 1) *2016*

Scholarship of Minglong Huang, USTC (top 10%) *2015*

National High School Mathematics Olympiad, Chinese Mathematics Society
The Second Prize in Province *2013*

PUBLICATIONS; LINK

J. Li, Z. Qu, and **M. Xu**, *Dynamic Balancing and Matchmaking in Competitive Live-Service Games*. Major Revision at Management Science (MS), 2026.

M. Xu and D. Klabjan, *Open Multi-agent Multi-armed Bandit with Applications in Permissionless Blockchain*. International Conference on Artificial Intelligence and Statistics (AISTATS), Tangier, Morocco, 2026.

W. Shen, Y. Zhang, M. Huang, **M. Xu**, J. Zhang, and C. Shen, *MLorc: Momentum Low-rank Compression for Large Language Model Adaptation*. International Conference on Artificial Intelligence and Statistics (AISTATS), Tangier, Morocco, 2026.

M. Xu, L. Shan, F. Ghaffari, X. Wang, X. Liu, and M. Hajiesmaili, *Heterogeneous Multi-agent Multi-armed Bandits on Stochastic Block Models*. Proceedings of the ACM on Measurement and Analysis of Computing Systems (ACM SIGMETRICS/POMACS), Ann Arbor, Michigan, 2026.

M. Xu and D. Klabjan, *Multi-agent Multi-armed Bandit Regret Complexity and Optimality*. International Conference on Artificial Intelligence and Statistics (AISTATS), Phuket, Thailand, 2025.

J. Liu, H. Qiu, L. Yang, and **M. Xu**, *Distributed Multi-agent Bandits Over Erdős–Rényi Random Networks*. Advances in Neural Information Processing Systems (NeurIPS), Vancouver, Canada, 2025.

X. Wang and **M. Xu**, *Multi-agent Multi-armed Bandit with Fully Heavy-tailed Dynamics*. Winter Simulation Conference (WSC), Seattle, Washington, 2025.

M. Xu and D. Klabjan, *Decentralized Blockchain-based Robust Multi-agent Multi-armed Bandit*. Coordination and Cooperation in Multi-Agent Reinforcement Learning (CoCoMARL; Honorable Mention), 2024.

M. Xu and D. Klabjan, *Decentralized Randomly Distributed Multi-agent Multi-armed Bandit with Heterogeneous Rewards*. Advances in Neural Information Processing Systems (NeurIPS Spotlight), New Orleans, Louisiana, 2023.

M. Xu and D. Klabjan, *Pareto Regret Analyses in Multi-objective Multi-armed Bandit*. International Conference on Machine Learning (ICML), Honolulu, Hawaii, 2023.

S. Wan, C. Zheng, Z. Sun, **M. Xu**, X. Yang, H. Zhu, and J. Guo, *GCF: Generalized Causal Forest for Heterogeneous Treatment Effect Estimation in Online Marketplace*. KDD Workshop on Decision Intelligence and Analytics for Online Marketplaces, Washington D.C., 2022.

PRESENTATIONS

Invited Presentation at **Next-Generation AI & Machine Learning**, Boston, MA *Nov. 2026*
Collective Intelligent Decision-Making Under Uncertainty via Multi-Agent Learning

Invited Presentation at **INFORMS**, San Francisco, CA *Nov. 2026*
Are Stochastic Multi-objective Bandits Harder than Single-objective Bandits?

Invited Presentation at **ACM SIGMETRICS**, Ann Arbor, MI *June. 2026*
Heterogeneous Multi-agent Multi-armed Bandits on Stochastic Block Models

Invited Presentation at **Mentoring Workshop**, Ann Arbor, MI *June. 2026*
Collective Intelligent Decision Making via Multi-agent Learning

Invited Presentation at **IISE**, Arlington, TX *May. 2026*
Multi-agent Bandits Over Erdős-Rényi and Heavy-tailed Random Networks

Invited Presentation at **UIUC**, Champaign, IL *Apr. 2026*
Multi-agent and Multi-objective Learning in Complex Systems

Invited Presentation at **Northwestern University**, Evanston, IL *Mar. 2026*
Alumni Seminar

Invited Presentation at **Mount Holyoke College**, Mount Holyoke, MA *Feb. 2026*
Multi-agent Bandits Over Erdős-Rényi and Heavy-tailed Random Networks

Invited Presentation at **DoorDash**, Seattle, WA *Dec. 2025*
Multi-agent Bandits Over Erdős-Rényi and Heavy-tailed Random Networks

Presentation at **WSC**, Seattle, WA *Dec. 2025*
Multi-agent Multi-armed Bandit with Fully Heavy-tailed Dynamics

Presentation at **NeurIPS**, San Diego, CA *Dec. 2025*
Distributed Multi-agent Bandits Over Erdos-Rényi Random Networks

Presentation at **INFORMS**, Atlanta, GA *Oct. 2025*
Multi-agent Bandits Over Erdős-Rényi and Heavy-tailed Random Networks

Invited Presentation at **Math Symposium in Association of Women in Math**, Amherst, MA *Apr. 2025*
Optimality and Complexity in Decentralized Multi-agent Multi-armed Bandits

Presentation at **Workshop in IMSI**, Chicago, IL *Mar. 2025*
Multi-agent Multi-armed Bandit with Fully Heavy-tailed Dynamics

Invited Presentation at **CS Theory Seminar**, Amherst, MA *Feb. 2025*
Optimality and Complexity in Decentralized Multi-agent Multi-armed Bandits

Presentation at **INFORMS 2024**, Seattle, WA *Oct. 2024*
Decentralized Blockchain-based Robust Multi-agent Multi-armed Bandit

Invited Presentation at **Northwestern University**, Virtual *Sep. 2024*
Alumni Seminar

Presentation at **RLC 2024**, Amherst, MA *Aug. 2024*
Decentralized Blockchain-based Robust Multi-agent Multi-armed Bandit

Presentation at **NeurIPS 2023**, New Orleans, LA *Dec. 2023*
Decentralized Randomly Distributed Multi-agent Multi-armed Bandit with Heterogeneous Rewards

Presentation at **INFORMS 2023**, Phoenix, AZ *Oct. 2023*
Multi-agent and Multi-objective Multi-armed Bandit

Invited Presentation at **LinkedIn**, Sunnyvale, California *Sep. 2023*
On natural language understanding and large language models for LinkedIn search recommendation

Presentation at **ICML 2023**, Honolulu, Hawaii *Jul. 2023*
Pareto Regret Analyses in Multi-objective Multi-armed Bandits

Invited Presentation at **LinkedIn**, Virtual *Oct. 2022*
GCF: Generalized Causal Forest for Heterogeneous Treatment Effect Estimation in Online Marketplace

Presentation at **KDD 2022**, Washington DC *Aug. 2022*
GCF: Generalized Causal Forest for Heterogeneous Treatment Effect Estimation in Online Marketplace

TEACHING EXPERIENCE

Instructor at UMass Amherst
- MIE 524: Machine Learning for Dynamic Decision Making (Spring 2027)
- MIE 373: Introduction to Simulation (Spring 2025, 2026)
- MIE 621: Descriptive Analytics (Fall 2024, 2025, 2026)

Teaching Assistant at Northwestern University
- MLDS-400: Everything starts with data (Fall 2023)
- IEMS-404/MLDS-401: Predictive analytics (Fall 2023)

Teaching Assistant at University of Science and Technology of China
- Real analysis (Spring 2017)
- Functions of complex variables (Fall 2016)

STUDENTS

Phd advisees:
Kenneth Chen (M.S. at University of Chicago)

Jingqi Fan (co-advised with Prof. Hajiesmaili at CICS, B.S. at Northeastern)

Zirui Liu (co-advised with Prof. Ma at CICS, M.S. at University of California Irvine)

Bingnan Huo (M.S. at Brown University)

Undergraduate project advisees:
Jingyuan Liu, Oct. 2024 - Apr. 2026 (at Nanjing University)

- now graduate student at UC Berkeley

John Wang, Sept. 2025 - Jun. 2026 (at UMass Amherst)

- now graduate student at UMass

Jose Cruz, Dec. 2024 - July 2025 (at UMass Amherst)

- now graduate student at UMass

Jingshuai Zhang, Sep. 2024 - July 2025 (at University of Minnesota - Twin Cities)

- now graduate student at UPenn

Jessica Lanin, Spring 2023 (at Northwestern)

Alejandra Bustamante, Spring 2023 (at Northwestern)

Tom Muma Ratemo, Spring 2023 (at Northwestern)

Tyler Shannon, Spring 2023 (at Northwestern)

Konyin Rilwan Okubadejo, Spring 2023 (at Northwestern)

PROFESSIONAL ACTIVITIES

Track Co-chair/organizer:
• IISE Annual Meeting & Conference, 2026

Session Chair:
• INFORMS Annual Meeting, 2026, 2025, 2024

Program Committee:
• Reviewer - AISTATS, NeurIPS, ICML, ICLR, ACML
• PC - ACM SIGMETRICS, International Workshop on Large Language Model Supply Chain Analysis (LLMSC 2025)

Journal Reviewer:
• INFORMS Journal on Computing
• TMLR
• OMEGA
• IISE
• IEEE TON

PhD Committee:
• Bingnan Huo, PhD student, Industrial Engineering at UMass Amherst
• Zirui Liu, PhD student, CICS at UMass Amherst
• Kelvin Ekun R., PhD student, Industrial Engineering at UMass Amherst
• Sonza Singh, PhD student, Industrial Engineering at UMass Amherst
• JaEon Cho, PhD student, Biomedical Engineering at UMass Amherst
• Amirmahdi Mirfakhar, PhD student, CICS at UMass Amherst
• Fatemeh Ghaffari, PhD student, CICS at UMass Amherst

Master Committee:
• Stephen Onyeukwu, PhD student, Industrial Engineering at UMass Amherst

Faculty Advisor:
• Undergraduate Club at UMass Amherst for Institute of Industrial and Systems Engineers (IISE), 2024-2025, 2025-2026
• Client Project Challenge (CPC) at Northwestern University, Spring 2024

Judge:
• Research Symposium, College of Engineering at UMass Amherst, 2025

Panelist:
• Data Science Night, Northwestern University, October 2024
• Academia Job Search, Northwestern University, April 2024